Result 1 (selected element identity). [Clean, selected element identity]
Let A be a type. Let S be a subset of A. Let x be a selected element of S. Then

$$x = x.$$

Result 2 (selected pair identity). [Clean, selected pair identity]
Let A be a type. Let S be a subset of A. Let x and y be selected elements of S. Then

$$x = x \text{ and } y = y.$$

Result 3 (eligible element identity). [Clean, eligible element identity]
Let A be a type. Let S be a subset of A. Let x be an eligible element of S. Then

$$x = x.$$

LLM-assisted statements and proofs (gpt-6, 2026-11-07). See PROVENANCE.md. Selected and eligible elements have the same Lean subtype.